BATCH DATA ENGINEERING PROJECT REPORT

Smart Patient Readmission Risk Analytics

End-to-End Batch Data Engineering & Clinical Risk Analytics Pipeline

Python • PySpark • Databricks • Delta Lake • SQL • AI/BI Dashboard

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Project	Smart Patient Readmission Risk Analytics

SMART PATIENT READMISSION RISK ANALYTICS

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Report scope

This report documents the complete Smart Patient Readmission Risk Analytics pipeline: raw hospital-style data generation, Bronze ingestion, Silver cleaning and feature engineering, Gold business aggregations, and the Databricks Executive Overview dashboard.

Abstract

The Smart Patient Readmission Risk Analytics project is a batch data engineering pipeline built on Databricks using the Medallion Architecture. The pipeline transforms raw patient, diagnosis, and hospital admission data into cleaned, enriched, and analytics-ready datasets that support readmission-risk analysis.

The Bronze layer preserves raw source data, including simulated data-quality issues, while the Silver layer performs cleaning, joins, standardization, and feature engineering. Business-oriented Gold tables are then generated to answer questions related to diagnosis-level readmissions, department performance, age-group risk, and individual patient risk.

The final analytics layer is presented through a Databricks Executive Overview dashboard. The dashboard summarizes 199 patients, 692 admissions, 153 readmissions, an overall readmission rate of 22.11%, and an average length of stay of 5.78 days. It also provides trend, department, diagnosis, age-group, risk-distribution, and high-risk patient views.

Core problem addressed

Hospitals need a repeatable analytical pipeline that can convert admission history into actionable indicators of 30-day readmission risk, while preserving data lineage and supporting operational decision-making.

Project Objectives

- Build a complete batch data engineering pipeline from raw hospital-style data to business insights.
- Implement the Medallion Architecture using Bronze, Silver, and Gold layers on Databricks.
- Preserve raw data in the Bronze layer for traceability and reproducibility.
- Clean and standardize admission data, including dates, categorical values, missing values, and duplicate records.
- Join patient and diagnosis information with admission events to create an enriched Silver dataset.
- Engineer analytical features such as age group, admission month, prior admissions, comorbidity proxy, readmission flag, and length of stay.
- Generate business-ready Gold tables for diagnosis, department, age-group, and patient-level risk analysis.
- Create an interactive Databricks dashboard for executive and operational readmission-risk monitoring.
- Validate row counts, schemas, and Gold-layer outputs before dashboard consumption.
- Demonstrate production-oriented Spark practices such as idempotent Delta writes, broadcast joins, AQE, and post-write optimization.

Technology Stack

The project combines distributed data processing, reliable Delta storage, SQL analytics, and dashboarding within Databricks.

Technology	Purpose
Python	Core scripting, synthetic data generation, and pipeline logic
PySpark	Distributed transformations, joins, aggregations, and feature engineering
Databricks	Cloud development environment, notebooks, SQL, and dashboarding
Delta Lake	Reliable, versioned storage for Bronze, Silver, and Gold tables
SQL	Business analytics, validation, and dashboard-ready queries
Unity Catalog	Centralized catalog and schema organization for project tables
Spark AQE	Adaptive query optimization and partition handling

Pipeline Execution Order

The notebooks are designed to be re-runnable and organized by the Medallion layers.

Step	Module	Purpose
1	Bronze / generate_data	Generate synthetic patient, diagnosis, and admission records and write raw Delta tables.
2	Silver / silver_transform	Clean, join, standardize, and engineer features into silver_admissions_enriched.
3	Gold / gold_aggregations	Compute four business aggregation tables for risk and performance analysis.
4	Analytics / SQL	Query Gold outputs and build the Databricks Executive Overview dashboard.

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Medallion Architecture

The pipeline progressively refines hospital admission data from raw ingestion to business-ready risk analytics.

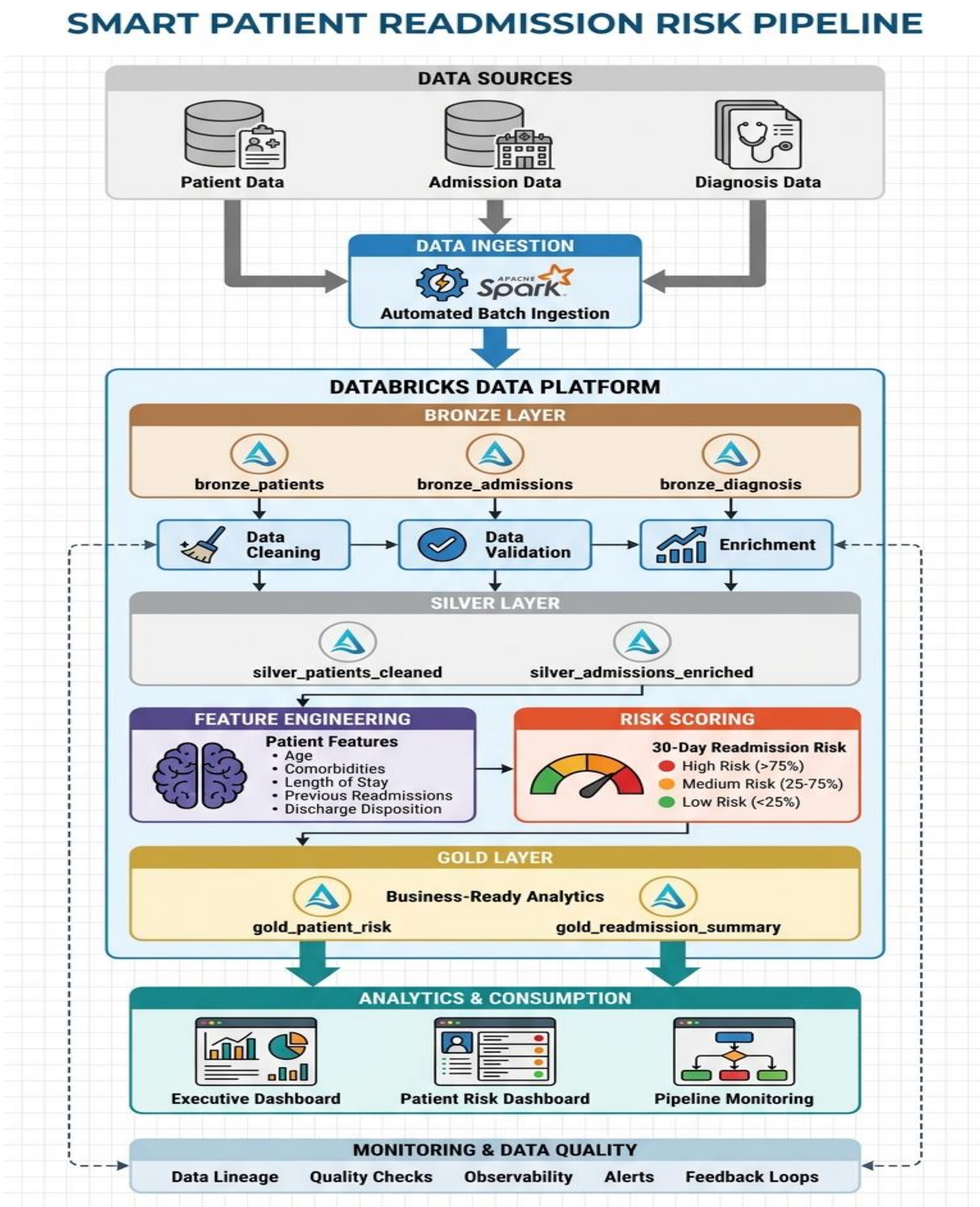

Fig1:Readmission Risk Pipeline architecture

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Architecture & Layer Details

BRONZE LAYER	SILVER LAYER	GOLD LAYER
Raw patient, diagnosis, and admission records • Original values preserved • Intentional data-quality issues retained • Delta tables for traceability	Duplicate removal and null handling • Categorical and date standardization • Patient/diagnosis/admission joins • Age group and admission month • Prior admission count • Comorbidity proxy • Recomputed length of stay	Readmission by diagnosis • Department performance • Age-group risk • Patient risk profile • Dashboard-ready business metrics

Analytics & consumption

Gold outputs feed the Databricks Executive Overview, where KPIs and visualizations expose readmission patterns by department, diagnosis, month, age group, and patient risk category.

Data Cleaning & Validation Report

Activity	Status	Validation Evidence
Raw patient DataFrame created	**Completed**	205 rows shown in Bronze validation
Admissions DataFrame created	**Completed**	692 rows shown in Bronze validation
Explicit schemas applied	**Completed**	Patient and admission fields typed explicitly
Silver enriched table created	**Completed**	Joined/cleaned dataset with 20 columns
Gold diagnosis table generated	**Completed**	9 diagnosis categories
Gold department table generated	**Completed**	7 departments
Gold age-group table generated	**Completed**	5 age groups
Gold patient risk table generated	**Completed**	199 patient profiles

Data quality approach

The project intentionally simulates realistic Bronze-layer issues such as nulls, inconsistent categorical values, and date noise. Silver transformations standardize the data before business aggregation, while validation cells confirm row counts and output availability.

Figure 2. Example Bronze patient DataFrame validation.

Gold Layer Tables

Four business-ready Gold tables form the analytical layer used by the dashboard.

Gold Table	Business Question	Key Metric
readmission_by_diagnosis	Which diagnosis categories have the highest readmission rates?	readmission_rate_pct
department_performance	Which departments show higher readmission performance risk?	readmission_rate_pct / performance metrics
age_group_risk	Which age groups have the highest readmission risk?	readmission_rate_pct
patient_risk_profile	Which individual patients need higher-priority attention?	risk_category / risk_score

Risk Categorization

- High Risk: patients with 3 or more readmissions in the risk-profile logic.
- Medium Risk: patients with 1–2 readmissions.
- Low Risk: patients with no readmissions.

Gold validation

Final validation confirmed 9 diagnosis rows, 7 department rows, 5 age-group rows, and 199 patient-risk-profile rows.

Dashboard Overview

The Databricks Executive Overview connects the Gold layer to an interactive management view.

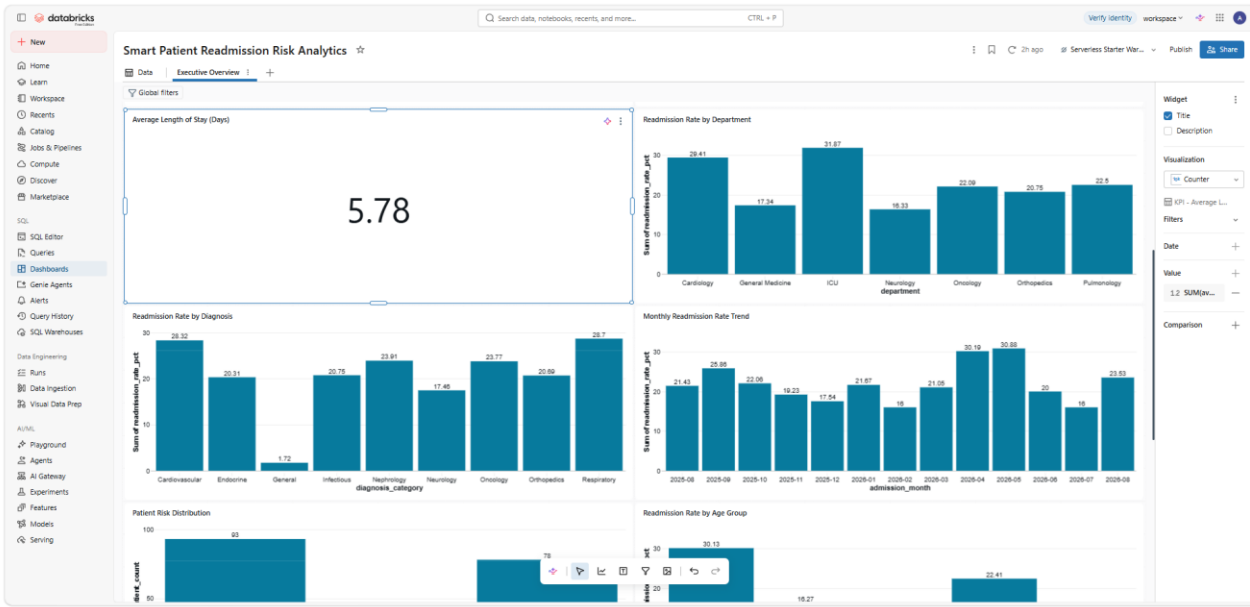

Figure 3. Smart Patient Readmission Risk Analytics – Executive Overview.

The dashboard includes KPI cards for total patients, admissions, readmissions, overall readmission rate, and average length of stay. Supporting charts show department performance, diagnosis-level readmission rates, monthly trends, patient risk distribution, age-group risk, and high-risk patient profiles.

Key Business Insights

Overall readmission	153 readmissions were recorded across 692 admissions, producing an overall readmission rate of 22.11%.
Patient risk distribution	93 patients are categorized as High Risk, 78 as Medium Risk, and 28 as Low Risk.
Age-group risk	Elderly patients show the highest readmission rate at 30.13%, followed by Seniors at 22.41%.
Department risk	ICU has the highest department readmission rate at 31.87%, followed by Cardiology at 29.41%.
Diagnosis risk	Respiratory and Cardiovascular categories show the highest diagnosis-level rates at 28.70% and 28.32%, respectively.
Monthly trend	Readmission rate peaks in May 2026 at 30.88% and April 2026 at 30.19%; the lowest observed monthly rate is 16.00% in February and July 2026.
Length of stay	Average length of stay across the dashboard population is 5.78 days.

Decision-support interpretation

The dashboard highlights where follow-up resources may be prioritized: elderly/high-risk patients, ICU and Cardiology, and diagnosis categories with higher observed readmission rates.

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Challenges Faced

- Understanding and implementing the Medallion Architecture across multiple Databricks notebooks and Delta tables.
- Managing data-quality simulation while ensuring that the Silver layer produced consistent, analytics-ready records.
- Handling joins between patient, diagnosis, and admission datasets without losing referential consistency.
- Designing derived features such as prior admission count, admission month, age group, and length of stay.
- Building multiple Gold aggregations and ensuring their row counts and schemas matched the intended business questions.
- Configuring dashboard visualizations correctly so that categorical dimensions and percentage measures were aggregated in a meaningful way.
- Validating each layer through Spark DataFrame outputs and final Gold-layer checks before publishing the dashboard.

Key engineering lesson

The project reinforced the importance of validating every layer independently instead of relying only on the final dashboard output.

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Learning Outcomes

- Gained practical experience with Databricks notebooks and the Lakehouse workflow.
- Learned how to structure a complete Bronze → Silver → Gold batch pipeline.
- Improved PySpark skills for DataFrame creation, joins, aggregations, window logic, and feature engineering.
- Learned how Delta tables support reliable, repeatable data pipelines.
- Practiced data-quality validation using row counts, schema inspection, null checks, and business-level verification.
- Learned how to turn raw admission events into clinically meaningful risk indicators.
- Developed experience designing Gold-layer tables around specific business questions.
- Learned how to build an Executive Overview dashboard directly from Databricks analytical outputs.
- Gained a stronger understanding of production-oriented Spark optimization, including AQE, broadcast joins, OPTIMIZE, ZORDER, and ANALYZE TABLE.

Future enhancement direction

The project can be extended with a machine-learning readmission prediction model, automated high-risk alerts, incremental MERGE processing, streaming ingestion, stronger data-quality expectations, and CI/CD deployment.

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Conclusion

The Smart Patient Readmission Risk Analytics pipeline successfully demonstrates an end-to-end data engineering workflow for hospital readmission analysis. Raw patient, diagnosis, and admission data are ingested into the Bronze layer, transformed and enriched in the Silver layer, and converted into business-ready Gold tables.

The final Databricks dashboard provides a concise view of the most important indicators: 199 patients, 692 admissions, 153 readmissions, a 22.11% overall readmission rate, and a 5.78-day average length of stay. The analytical views also identify higher-risk age groups, departments, diagnosis categories, monthly periods, and individual patient profiles.

Overall, the project demonstrates how a structured Medallion Architecture can convert raw healthcare-style operational data into reliable analytics and decision-support information. It provides a strong foundation for future predictive modeling, automated alerts, and scalable clinical analytics.

199 Patients	**692** Admissions	**153** Readmissions	**22.11%** Overall Rate